

Reproduction of nerfstudio-project/gsplat: 3D Gaussian Splatting on Mip-NeRF 360 garden

K-Veritas Team

Submitted by K-Veritas Team

kv:2610.00022v1 · Published 2026-10-08

Independent reproduction · Artifact evaluation · Computer vision

Abstract

Trained 3D Gaussian Splatting with gsplat's simple_trainer.py default strategy at the pinned commit on the Mip-NeRF 360 garden scene at quarter resolution, 30,000 steps, with the built-in held-out evaluation (PSNR, SSIM, LPIPS) at 7,000 and 30,000 steps. gsplat and its CUDA dependencies were compiled for sm_120 with CUDA 12.8 and PyTorch 2.9.1 as pinned. One NVIDIA RTX 5060 Ti.

About this record

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1 Sealed reports

The experiments were sealed with K-Veritas [1], which binds each result to the code, hardware and time that produced it.

Report	Label	Sealed	Runs	Invocations	Signer
r1 [2]	Mip-NeRF 360 garden, 30k st...	2026-10-08	1	1	Signed by the K-Veritas server

2 Results

r1: Mip-NeRF 360 garden, 30k steps

Run	Metric	Final value	Step
1	psnr	26.6525	7000
1	ssim	0.838914	7000
1	lpips	0.117377	7000
1	num_GS	3.98882e+06	7000

Run	Inline claim	Value	Phase
1	psnr	26.6525	
1	ssim	0.838914	
1	lpips	0.117377	

3 Setup

Report	Item	Value
r1	Operating system	linux / amd64
r1	CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
r1	GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
r1	Memory	31.0 GB
r1	Total run time	2334.4 s

4 Files

Each report and code bundle must hash to these values. A bundle belongs to the report whose signed data names its hash.

File	SHA-256	Size
r1.pdf	541d85b49f8a9efd59cbc8d1f40f5d4796e98259acb4370940b19352caa28811	7.6 MB
r1.kvbundle.zip	1d3c2150d038e0cfd3a89afb1e02174ebcc02418e9132d4de95841a7d5a86fe7	2.5 MB

5 Links and licence

Paper: <https://arxiv.org/abs/2409.06765>

Reproduction of: <https://github.com/nerfstudio-project/gsplat/tree/512d366b67073d77ca099ede742683c165dfc23b>

Record: CC-BY 4.0. Code: Apache-2.0, as declared by the authors. Tags: gaussian-splatting, novel-view-synthesis, 3d-vision, mip-nerf-360.

Listed, not endorsed. Review checks the artifact, never the science. Reports are never edited; changing them makes a new version. Other fields are corrected in place, and every correction is logged in metadata.yaml.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

[2] Sealed experiment report r1 (Mip-NeRF 360 garden, 30k steps), kv:2610.00022v1/r1. Data hash 672f2745529beaea73a6011458121c12a7b04be31c78bd252618517f19f68aa1.
<https://kveritas.org/records/2610.00022v1/r1.pdf>